Global well-posedness and optimal time decay rates of solutions to the (pressureless) Euler-Navier-Stokes system

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Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory
April 4, 2024, Seoul

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5. Furture works

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Background

The coupled pressureless Euler-Navier-Stokes system reads as

$$\begin{cases} \partial_t \rho + \mathrm{div}(\rho u) = 0, \\ \partial_t(\rho u) + \mathrm{div}(\rho u \otimes u) = -\rho(u - v), \\ \partial_t v + v \cdot \nabla v + \nabla P = \Delta v + \rho(u - v), \\ \mathrm{div}_x v = 0, \end{cases} \tag{1}$$

with the initial data

$$(\rho, u, v)|_{t=0} = (\rho_0, u_0, v_0). \tag{2}$$

The $\rho = \rho(t,x)$ and $u = u(t,x)$ denote the density and velocity of the pressureless Euler fluid flow; $v = v(t,x)$ and $P = P(t,x)$ represent the velocity and pressure of the incompressible Navier-Stokes fluid flow.

Background

The coupled (pressureless) Euler-Navier-Stokes system can be formally derived from the Vlasov(-Fokker-Planck)/incompressible Navier-Stokes equations, which describes the behavior of a large cloud of particles interacting with the incompressible fluid.

$$
\begin{cases}
\partial_t f + \xi \cdot \nabla_x f + \mathrm{div}_\xi((v-\xi)f) = -\alpha \mathrm{div}_\xi((u_f-\xi)f) + \sigma \Delta_\xi f, \\
\partial_t v + v \cdot \nabla_x v + \nabla_x P = \Delta_x v + \int_{\mathbb{R}^n} (\xi - v) f d\xi, \\
\mathrm{div}_x v = 0,
\end{cases} \tag{3}
$$

for $(t,x,\xi) \in \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^n$, where u_f denotes the averaged local velocity defined by

$$
u_f = u_f(t,x) = \frac{\int_{\mathbb{R}^n} \xi f(t,x,\xi) d\xi}{\int_{\mathbb{R}^n} f(t,x,\xi) d\xi}. \tag{4}
$$

Background

- $\alpha = 0$, the kinetic-fluid model reduces to the Vlasov-Navier-Stokes-Fokker-Planck equations.
- $\sigma = 0$, the kinetic-fluid model reduces to the Vlasov-Navier-Stokes system with a local alignment force.
- $\alpha = 0, \sigma = 0$, the kinetic-fluid model reduces to the Vlasov-Navier-Stokes system.

Background

- $\alpha = \sigma = \epsilon^{-1}$.

Let $(f^\epsilon, v^\epsilon, P^\epsilon)$ be the solution to the system (3) with $\alpha = \sigma = \epsilon^{-1}$. Then we have

$$\begin{cases} \partial_t f^\epsilon + \xi \cdot \nabla_x f^\epsilon + \mathrm{div}_\xi((v-\xi)f^\epsilon) = -\frac{1}{\epsilon}\mathrm{div}_\xi((u_{f^\epsilon}-\xi)f^\epsilon) + \frac{1}{\epsilon}\Delta_\xi f^\epsilon, \\ \partial_t v^\epsilon + v^\epsilon \cdot \nabla_x v^\epsilon + \nabla_x P^\epsilon = \Delta_x v^\epsilon + \int_{\mathbb{R}^n} (\xi - v^\epsilon) f^\epsilon d\xi, \\ \mathrm{div}_x v^\epsilon = 0. \end{cases} \tag{5}$$

In terms of $(5)_1$, we verify that

$$-\mathrm{div}_\xi((u_{f^\epsilon}-\xi)f^\epsilon) + \Delta_\xi f^\epsilon \to 0, \quad \text{as} \quad \epsilon \to 0.$$

Thus, the particle distribution function $f^\epsilon(t,x,\xi)$ is expected to converge to

$$f^\epsilon(t,x,\xi) = \frac{\rho_f(t,x)}{(2\pi)^{n/2}} e^{-\frac{|u_f-\xi|^2}{2}}, \quad \text{and} \quad \rho_f(t,x) = \int_{\mathbb{R}^n} f^\epsilon(t,x,\xi) d\xi. \tag{6}$$

Integrating $(5)_1$ with respect to ξ over $\mathbb{R}^n$, we can easily obtain the continuity equation

$$\partial_t \rho_f + \mathrm{div}_x(\rho_f u_f) = 0. \tag{7}$$

Multiplying $(5)_1$ by ξ and integrating equation with respect to ξ in $\mathbb{R}^n$ yield

$$\begin{aligned} &\frac{d}{dt}\int_{\mathbb{R}^n} \xi f^\epsilon(t,x,\xi)d\xi \\ &= \cdots \\ &= -\mathrm{div}_x\Big(\int_{\mathbb{R}^n} \xi \otimes \xi f^\epsilon d\xi\Big) + \int_{\mathbb{R}^n}(v^\epsilon - \xi)f^\epsilon d\xi \\ &\triangleq I_1^\epsilon + I_2^\epsilon, \end{aligned} \tag{8}$$

Background

The first term on the right-hand side in (8) is stated as

$$I_1^\epsilon = -\mathrm{div}_x\Big(\int_{\mathbb{R}^n} (\xi - u_f) \otimes (\xi - u_f) d\xi + 2\int_{\mathbb{R}^n} \xi \otimes u_f f^\epsilon d\xi - \int_{\mathbb{R}^n} u_f \otimes u_f f^\epsilon d\xi \Big). \tag{9}$$

By (4), the second term is calculated as

$$I_2^\epsilon = \rho_f(v^\epsilon - u_f).$$

Assuming $v^\epsilon \to v, P^\epsilon \to P$ as $\epsilon \to 0$, we obtain

$$\lim_{\epsilon\to 0} \frac{d}{dt}\int_{\mathbb{R}^n} \xi f^\epsilon d\xi = \frac{d}{dt}\int_{\mathbb{R}^n} \xi f d\xi = \frac{d}{dt}(\rho_f u_f), \tag{10}$$

$$\begin{aligned} I_1 = \lim_{\epsilon\to 0} I_1^\epsilon &= -\mathrm{div}_x\Big(\int_{\mathbb{R}^n} (\xi - u_f) \otimes (\xi - u_f) \frac{\rho_f}{(2\pi)^{n/2}} e^{-\frac{|u_f-\xi|^2}{2}} d\xi \\ &+ 2\int_{\mathbb{R}^n} \xi \otimes u_f f d\xi - \int_{\mathbb{R}^n} u_f \otimes u_f \frac{\rho_f}{(2\pi)^{3/2}} e^{-\frac{|u_f-\xi|^2}{2}} d\xi \Big) \\ &= -\mathrm{div}_x(\rho_f \mathrm{I}_3 + 2\rho_f u_f \otimes u_f - \rho_f u_f \otimes u_f) \\ &= -\nabla_x \rho_f - \mathrm{div}_x(\rho_f u_f \otimes u_f), \end{aligned} \tag{11}$$

We follows from above to obtain

$$
\begin{cases}
\partial_t \rho_f + \mathrm{div}_x(\rho_f u_f) = 0, \\
\partial_t(\rho_f u_f) + \mathrm{div}_x(\rho_f u_f \otimes u_f) + \nabla_x \rho_f = -\rho_f(u_f - v), \\
\partial_t v + v \cdot \nabla_x v + \nabla_x P = \Delta_x v + \rho_f(u_f - v), \\
\mathrm{div}_x v = 0.
\end{cases} \tag{12}
$$

Background

- $\alpha = \epsilon^{-1}, \sigma = 0.$

Then we have

$$\begin{cases} \partial_t f^\epsilon + \xi \cdot \nabla_x f^\epsilon + \mathrm{div}_\xi((v-\xi)f^\epsilon) = -\frac{1}{\epsilon}\mathrm{div}_\xi((u_f - \xi)f^\epsilon), \\ \partial_t v^\epsilon + v^\epsilon \cdot \nabla_x v^\epsilon + \nabla_x P^\epsilon = \Delta_x v^\epsilon + \int_{\mathbb{R}^3} (\xi - v^\epsilon) f^\epsilon d\xi, \\ \mathrm{div}_x v^\epsilon = 0. \end{cases} \tag{13}$$

Thus, the particle distribution function $f^\epsilon(t,x,\xi)$ is expected to converge to

$$f^\varepsilon(t,x,\xi)dxd\xi \to \rho^\varepsilon(t,x)dx \otimes \delta_{u^\varepsilon(x,t)}(d\xi). \tag{14}$$

Background

Similarly, we can obtain

$$\begin{cases} \partial_t \rho_f + \mathrm{div}_x(\rho_f u_f) = 0, \\ \partial_t(\rho_f u_f) + \mathrm{div}_x(\rho_f u_f \otimes u_f) = -\rho_f(u_f - v), \\ \partial_t v + v \cdot \nabla_x v + \nabla_x P = \Delta_x v + \rho_f(u_f - v), \\ \mathrm{div}_x v = 0. \end{cases} \tag{15}$$

Known Results

Incompressible Navier-Stokes equations,

- Leray [1934,Acta Math.] global well-posedness; Fujita-Kato [1964,ARMA.] well-posedness of $\dot{H}^{1/2}$;Cannone[1997,Rev.Math.Iberoam]global in Besov Space;Planchon[1998,Rev.Math.Iberoam] Asymptatic behavior of strong solution, Koch–Tataru[2001,Adv.]global-in-time solutions in $BMO^{-1}\cdots$.
- M. Schonbek. [1985,ARMA;1991,JAMS]. Algebraic decay rate.
- J. C. Robinson, J. L. Rodrigo, and W. Sadowski [2006,Cambridge University Press]. The Three-Dimensional Navier-Stokes Equations.

Known Results

For pressureless Euler equations,

- Brenier, Y. and Grenier, E.(1998,SIAM), Engelberg, S.(1996,Physica.D),E, W., Rykov,Y. and Sinai, Y.(1996,CMP),Huang, F. and Wang, Z.(2001,CMP) proved pressureless Euler equations develop a finite-time formation of singularities, for instance, δ-shock.

For Euler with compressible Navier-Stokes equations,

- Choi(2016,SIAM) proved the global well-posedness and exponential time decay rates of the classical solution in the periodic domain with compressible Navier-Stokes equations.
- Wu-Zhang-Zou(2020,SIAM) obtained the large-time behavior in $\mathbb{R}^3$.
- Wu-Zou(2021) studied the pointwise space-time behavior of the Cauchy problem to this system.

For pressureless Euler with incompressible Navier-Stokes equations,

- Choi and Jung[JMFM,2021] proved the global well-posedness of system in the whole space.

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Motivation

The coupled pressureless Euler-Navier-Stokes system reads as

$$\begin{cases} \partial_t \rho + \mathrm{div}(\rho u) = 0, \\ \partial_t(\rho u) + \mathrm{div}(\rho u \otimes u) = -\rho(u - v), \\ \partial_t v + v \cdot \nabla v + \nabla P = \Delta v + \rho(u - v), \\ \mathrm{div} v = 0. \end{cases} \tag{16}$$

Pressureless Euler equations: Global solutions doesn't exist.
Navier-Stokes equations: Global solutions exist for small initial data.
Q: How about the Euler-Navier-Stokes system?

Main results

Theorem 2.1 (Huang-Tang-Zou,2023,arXiv:2307.11581)

Let the integers $n \geq 3$ *and* $s \geq 2[\frac{n}{2}]+1$. *Assume the initial data satisfy* $\rho_0(x) > 0$, $\rho_0(x) \in H^s(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, $u_0 \in H^{s+2}(\mathbb{R}^n)$, $v_0 \in H^{s+1}(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$ *and* $\operatorname{div} v_0 = 0$. *There exists a sufficiently small constant* $\delta_0 > 0$ *such that if*

$$\|\rho_0\|_{H^s(\mathbb{R}^n)} + \|u_0\|_{H^{s+2}(\mathbb{R}^n)} + \|v_0\|_{H^{s+1}(\mathbb{R}^n)} \leq \delta_0, \tag{17}$$

then the pressureless E-NS system (118)-(2) *admits a global and unique classical solution for all* $(t,x) \in [0,+\infty) \times \mathbb{R}^n$ *and*

$$\sup_{t\geq 0}(\|\rho(t,x)\|_{H^s((\mathbb{R}^n))} + \|u(t,x)\|_{H^{s+2}(\mathbb{R}^n)} + \|v(t,x)\|_{H^{s+1}(\mathbb{R}^n)}) \leq C\delta_0. \tag{18}$$

Meanwhile, it holds for $t \geq 0$ *that*

$$\|u(t)\|_{H^{s+1}(\mathbb{R}^n)} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \quad \|v(t)\|_{H^{s+1}(\mathbb{R}^n)} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \tag{19}$$

where $\mathcal{I}_0 > 0$ is given by

$$\mathcal{I}_0 = \delta_0 + \|\rho_0\|_{L^1} + \|v_0\|_{L^1}. \tag{20}$$

Remark 1

Choi and Jung[JMFM,2021] proved the global well-posedness of system (118)-(2) in the whole space and obtained

$$\|u(t)\|_{H^{s+1}(\mathbb{R}^n)} + \|v(t)\|_{H^s(\mathbb{R}^n)} \le C(1+t)^{-\alpha}, \quad \forall\ 0 < \alpha < \frac{n}{4}, \tag{21}$$

under the assumption that

$$\|\rho_0\|_{H^s(\mathbb{R}^n)} + \|u_0\|_{H^{s+2}(\mathbb{R}^n)} + \|v_0\|_{H^{s+1}(\mathbb{R}^n)} + \|v_0\|_{L^1(\mathbb{R}^n)} \ll 1, \quad \rho_0(x) > 0. \tag{22}$$

In Theorem 2.1, the smallness condition for $\|v_0\|_{L^1}$ in (22) is removed and the decay rate for $\|\nabla^{s+1} v\|_{L^2}$ is further derived in (19) while only $\|\nabla^s v\|_{L^2}$ is obtained in (21).

The lower bound of $\|v\|_{L^2}$ is achieved as the same order as the upper one given in (19) by selecting special initial data.

Theorem 2.2 (Huang-Tang-Zou,2023,arXiv:2307.11581)

Assume the same conditions in Theorem 2.1 hold. There exists a sufficiently small constant $\delta_0 > 0$ such that if

$$\|\rho_0\|_{H^s(\mathbb{R}^n)} + \|u_0\|_{H^{s+2}(\mathbb{R}^n)} + \|v_0\|_{H^{s+1}(\mathbb{R}^n)} + \|v_0\|_{L^1(\mathbb{R}^n)} \le \delta_0, \tag{23}$$

and the Fourier transform of the initial velocity $\hat{v}_0(\xi)$ satisfies

$$\inf_{|\xi|\le 1} |\hat{v}_0(\xi)| \ge \delta_0^{\frac{3}{2}}, \tag{24}$$

then it holds for large time that

$$\frac{1}{2}\delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}} \le \|v(t)\|_{L^2(\mathbb{R}^n)} \le C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \tag{25}$$

Main results

Remark 2

From (25), the decay rate of $\|v\|_{L^2}$ obtained in Theorems 2.1 and 2.2 is optimal.

Remark 3

In the special case of $\rho = 0$, Theorem 2.2 coincides with the results M. Schonbek.[1991,JAMS].

Remark 4

Due to the absence of the pressure term in the Euler equations on (118), it is not expected to obtain the time decay rate of the fluid density $\rho(t, x)$.

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Local-time existence and uniqueness of classical solutions

Theorem 3.1 (Ha-2014-MMMAS,Choi-2016-JDE)

Let $n \geq 3$ *and* $s \geq 2[\frac{n}{2}] + 1$. *Assume the initial data satisfy* $\rho_0(x) > 0$, $\rho_0(x) \in H^s(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, $u_0 \in H^{s+2}(\mathbb{R}^n)$, $v_0 \in H^{s+1}(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$ *and* $\mathrm{div}\, v_0 = 0$. *There exists a small positive constant* $\epsilon_0 < \delta_0$ *such that if*

$$\|\rho_0\|_{H^s} + \|u_0\|_{H^{s+2}} + \|v_0\|_{H^{s+1}} \leq \epsilon_0, \tag{26}$$

then the pressureless E-NS system (118)-(2) *has a unique solution* (ρ, u, v),

$$(\rho, u, v) \in \mathcal{C}([0, T_0]; H^s(\mathbb{R}^n)) \times \mathcal{C}([0, T_0]; H^{s+2}(\mathbb{R}^n)) \times \mathcal{C}([0, T_0]; H^{s+1}(\mathbb{R}^n))$$

satisfying $\rho(t, x) > 0$ *for all* $(t, x) \in [0, T_0] \times \mathbb{R}^n$ *and*

$$\sup_{0 \leq t \leq T_0} \left(\|\rho(t, x)\|_{H^s} + \|u(t, x)\|_{H^{s+2}} + \|v(t, x)\|_{H^{s+1}}\right) \leq \delta_0,$$

where $T_0 > 0$ *is a fixed constant.*

A priori assumption

We extend the short-time classical solution to be a global one by establishing the uniform estimates. It is natural to provide the a priori assumption for sufficiently small $\delta > 0$,

$$\sup_{t \geq 0} \mathcal{Z}(t) \leq \delta, \quad \rho(t,x) > 0. \tag{27}$$

For notation simplicity, we denote the energy $\mathcal{Z}(t)$ and disspation $\mathcal{D}(t)$ below

$$\begin{aligned} \mathcal{Z}(t) &\triangleq \Big(\|\rho(t)\|_{H^s}^2 + \|u(t)\|_{H^{s+2}}^2 + \|v(t)\|_{H^{s+1}}^2\Big)^{\frac{1}{2}}, \\ \mathcal{D}(t) &\triangleq \Big(\|\nabla u(t)\|_{H^{s+1}}^2 + \|\nabla v(t)\|_{H^{s+1}}^2 + \|(u-v)(t)\|_{L^2}^2\Big)^{\frac{1}{2}}. \end{aligned} \tag{28}$$

Auxiliary lemmas

Lemma 1

(1) For any integer $k \geq 1$ and the pair of functions $f, g \in H^k(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, we have

$$\|\nabla^k(fg)\|_{L^2} \leq C\|f\|_{L^\infty}\|\nabla^k g\|_{L^2} + C\|\nabla^k f\|_{L^2}\|g\|_{L^\infty}. \tag{29}$$

Moreover if $\nabla f \in L^\infty(\mathbb{R}^n)$, we obtain

$$\left\|\nabla^k(fg) - f\nabla^k g\right\|_{L^2} \leq C\|\nabla f\|_{L^\infty}\|\nabla^{k-1} g\|_{L^2} + C\|\nabla^k f\|_{L^2}\|g\|_{L^\infty}. \tag{30}$$

(2) Let integers $n \geq 3$ and $s \geq \left[\frac{n}{2}\right] + 2$. If $f \in H^s(\mathbb{R}^n)$, then it holds the inequality

$$\|f\|_{L^\infty} \leq C\|\nabla f\|_{H^{s-2}}. \tag{31}$$

Auxiliary lemmas

Let $w(t,x)$ be the solution of the heat equation with initial data $w(0,x) = v_0$ in $(t,x) \in \mathbb{R}_+ \times \mathbb{R}^n$, which reads

$$\begin{cases} w_t - \Delta w = 0, \\ w(0,x) = v_0(x). \end{cases} \tag{32}$$

Lemma 2

Let integers $n \geq 3$ and $s \geq [\frac{n}{2}] + 2$. Assume $w(t,x)$ is the solution of (32), then it holds for $t \geq 0$ that

$$\|w(t,x)\|_{L^\infty} \leq C(1+t)^{-\frac{n}{2}}(\|v_0\|_{L^1} + \|\nabla v_0\|_{H^{s-2}}).$$

Lemma 3

Assume the Fourier transform of the initial velocity $\hat{v}_0(\xi)$ satisfies

$$\inf_{|\xi| \leq 1} |\hat{v}_0(\xi)| \geq \delta_0^{\frac{3}{2}}. \tag{33}$$

Then, it holds

$$\|w(t,x)\|_{L^2} \geq \delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}}. \tag{34}$$

Time decay rates of (u, v)

Proposition 1

Assume the conditions in Theorem 2.1 hold. Let $(\rho, u, v)(t, x)$ *be the classical solution of the system* (118)-(2) *satisfying the a priori assumption* (27)*. Then, it holds for* $t \geq 0$ *that*

$$\|u(t)\|_{L^2} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \quad \|v(t)\|_{L^2} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \tag{35}$$

where the positive constant C *is independent of time and* $\mathcal{I}_0$ *is given by* (20)*.*

Proof of Proposition 1

Proof.

Basic Energy conserved,

$$\frac{1}{2}\frac{d}{dt}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx+\|v\|_{L^2}^2\Big)dx+\int_{\mathbb{R}^n}\rho|u-v|^2dx+\|\nabla v\|_{L^2}^2=0. \tag{36}$$

After a direct calculation, by the a priori assumption (27), it is shown that

$$\begin{aligned}\int_{\mathbb{R}^n}\rho|u|^2dx+\|v\|_{L^2}^2 &\le 2\int_{\mathbb{R}^n}\rho|u-v|^2dx+2\int_{\mathbb{R}^n}\rho|v|^2dx+\|v\|_{L^2}^2\\ &\le 2\int_{\mathbb{R}^n}\rho|u-v|^2dx+(2\|\rho\|_{L^\infty}+1)\|v\|_{L^2}^2\\ &\le 2\int_{\mathbb{R}^n}\rho|u-v|^2dx+(2\delta+1)\|v\|_{L^2}^2\\ &\le 2(\int_{\mathbb{R}^n}\rho|u-v|^2dx+\|v\|_{L^2}^2),\end{aligned} \tag{37}$$

where we use the fact (31) and the smallness of δ. □

Proof of Proposition 1

Proof.

To obtain the time decay rates of $\|v\|_{L^2}$, we define the sets $X_1(t)$ and $X_1^c(t)$ as follows

$$X_1(t) = \left\{\xi : |\xi| \le \left(\frac{n}{t+n}\right)^{\frac{1}{2}}\right\},\ X_1^c(t) = \left\{\xi : |\xi| > \left(\frac{n}{t+n}\right)^{\frac{1}{2}}\right\}. \tag{38}$$

Then we are able to verify

$$\begin{aligned}&\frac{1}{2}\frac{d}{dt}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx + \|v\|_{L^2}^2\Big) + \int_{\mathbb{R}^n}\rho|u-v|^2dx\\ &+\int_{X_1(t)}|\xi|^2|\hat{v}(t,\xi)|^2d\xi + \int_{X_1^c(t)}|\xi|^2|\hat{v}(t,\xi)|^2d\xi = 0.\end{aligned} \tag{39}$$

which yields

$$\frac{d}{dt}E(t) + \frac{n}{t+n}E(t) \le 2n(1+t)^{-1}\int_{X_1(t)}|\hat{v}(t,\xi)|^2d\xi. \tag{40}$$

Proof of Proposition 1

Proof.

To estimate $\hat{v}(t,\xi)$, we rewrite $(118)_3$ as follows

$$v_t - \Delta v = F, \quad \text{and} \quad F \triangleq -v\cdot\nabla v - \nabla P + \rho(u-v). \tag{41}$$

Taking Fourier transform, we have by Duhamel's principle that

$$\hat{v}(t,\xi) = e^{-|\xi|^2 t}\hat{v}_0 + \int_0^t e^{-|\xi|^2(t-\tau)}\hat{F}(\tau,\xi)d\tau. \tag{42}$$

where

$$\nabla P = \cdots = \nabla(-\Delta)^{-1}\mathrm{div}\mathrm{div}(v\otimes v) - \nabla(-\Delta)^{-1}\mathrm{div}(\rho(u-v)).$$

□

Then the Fourier transform of F can be expressed as

$$|\hat{F}(\tau,\xi)| \le (2\pi)^{-\frac{n}{2}}|\xi|\|v\|_{L^2}^2 + (2\pi)^{-\frac{n}{2}}\|\rho(u-v)\|_{L^1} \le |\xi|\|v\|_{L^2}^2 + \|\rho(u-v)\|_{L^1}. \tag{43}$$

Proof of Proposition 1

Proof.

$$
\begin{aligned}
\int_0^t \|\rho(u-v)(\tau)\|_{L^1} d\tau &\le \int_0^t \|\rho\|_{L^1}^{\frac{1}{2}} \Big(\int_{\mathbb{R}^n} \rho|u-v|^2 dx\Big)^{\frac{1}{2}} d\tau \\
&\le \|\rho_0\|_{L^1}^{\frac{1}{2}} \int_0^t (1+\tau)^{-\frac{n+2}{8}} (1+\tau)^{\frac{n+2}{8}} \Big(\int_{\mathbb{R}^n} \rho|u-v|^2 dx\Big)^{\frac{1}{2}} d\tau \\
&\le \cdots \\
&\le 2\|\rho_0\|_{L^1}^{\frac{1}{2}} (10\epsilon_0 M(t) + 2\epsilon_1 E_0)^{\frac{1}{2}},
\end{aligned} \tag{44}
$$

Where

$$
\epsilon_0 \triangleq (1+t_0)^{-\frac{1}{4}}, \quad \epsilon_1 \triangleq (1+t_0)^{\frac{n+2}{4}}, \tag{45}
$$

$$
M(t) = \sup_{0\le\tau\le t} \Big\{(1+\tau)^{\frac{n}{2}} \Big(\int_{\mathbb{R}^n} \rho|u|^2 dx + \|v(\tau)\|_{L^2}^2\Big)\Big\}. \tag{46}
$$

□

Proof of Proposition 1

Proof.

Then, it holds

$$\int_{X_1(t)} |\hat{v}(t,\xi)|^2 d\xi \\ \leq C_1(1+t)^{-\frac{n}{2}}\left(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}(10\epsilon_0 M(t) + 2\epsilon_1 E_0)\right) + C_2(1+t)^{-\frac{n+2}{2}} M(t)^2. \tag{47}$$

□

The combination of (40) and (47) yields

$$\frac{d}{dt}E(t) + \frac{n}{t+n}E(t) \\ \leq 2nC_1(1+t)^{-\frac{n+2}{2}}\left(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}(10\epsilon_0 M(t) + 2\epsilon_1 E_0)\right) + 2nC_2(1+t)^{-\frac{n+4}{2}} M(t)^2. \tag{48}$$

Multiplying (48) by $(t+n)^n$, integrating with time give

$$E(t) \leq C_3(1+t)^{-\frac{n}{2}}\left(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}(10\epsilon_0 M(t) + 2\epsilon_1 E_0)\right) + C_4(1+t)^{-\frac{n+2}{2}} M(t)^2, \tag{49}$$

Proof of Proposition 1

Proof.

where C_3 and C_4 are positive constants independent of time and only depend on n. Choosing t_0 as,

$$t_0 = \max\left\{(40C_3\|\rho_0\|_{L^1})^4, 16C_3C_4(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}^2)\right\}. \tag{50}$$

Then, according to the definition of ϵ_0 and ϵ_1 in (45), we obtain

$$10\epsilon_0 C_3\|\rho_0\|_{L^1} \le \frac{1}{4}, \quad 4C_3C_4(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}^2)(1+t_0)^{-1} \le \frac{1}{4}, \quad 2\epsilon_1 E_0 \le \|\rho_0\|_{L^1}. \tag{51}$$

Due to the smallness of $M(0)$ and the continuity of $M(t)$, we can obtain

$$M(t) \le 4C_3(\|v_0\|_{L^1}^2 + \|\rho_0\|_{L^1}^2). \tag{52}$$

Then

$$\left(\int_{\mathbb{R}^n} \rho|u|^2 dx\right)^{\frac{1}{2}} \le C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \quad \|v(t)\|_{L^2} \le C\mathcal{I}_0(1+t)^{-\frac{n}{4}}. \tag{53}$$

Proof of Proposition 1

Proof.

Since we assume $\rho(t,x) > 0$, the second equation in (118) can be formulated to

$$u_t + u \cdot \nabla u + u - v = 0. \tag{54}$$

Integrating it in $\mathbb{R}^n$ yields

$$\frac{1}{2}\frac{d}{dt}\|u\|_{L^2}^2 + \|u\|_{L^2}^2 = \int_{\mathbb{R}^n} (-u \cdot \nabla u + v) \cdot u dx.$$

Then

$$\begin{aligned}
&\Big| \int_{\mathbb{R}^n} (-u \cdot \nabla u + v) \cdot u dx \Big| \\
&\leq \|u\|_{L^2} \|\nabla u\|_{L^\infty} \|u\|_{L^2} + \|v\|_{L^2} \|u\|_{L^2} \\
&\leq \cdots \\
&\leq C\delta \|u\|_{L^2}^2 + \frac{1}{4}\|u\|_{L^2}^2 + \|v\|_{L^2}^2 \\
&\leq \frac{1}{2}\|u\|_{L^2}^2 + C\mathcal{I}_0^2 (1+t)^{-\frac{n}{2}},
\end{aligned}$$

High-order derivatives of the velocities

Proposition 2

Assume the conditions in Theorem 2.1 hold. Let $(\rho, u, v)(t, x)$ *be the classical solution of the system* (118)-(2) *satisfying the a priori assumption* (27)*. Then, it holds for* $t \geq 0$ *that*

$$\|\nabla u(t)\|_{H^s} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \quad \|\nabla v(t)\|_{H^s} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \tag{55}$$

where the positive constant C *is independent of time.*

Proof.

Define $N(t)$ as follows

$$N(t)=\sup_{0\leq\tau\leq t}\left\{(1+\tau)^{\frac{n}{4}}(\|\nabla u(\tau)\|_{H^s}+\|\nabla v(\tau)\|_{H^s})\right\}, \tag{56}$$

with the integer $s\geq 2[\frac{n}{2}]+1$. For $1\leq k\leq s+1$, applying ∇^k to $(118)_3$ and taking the L^2 inner product with $\nabla^k v$ in $\mathbb{R}^n$ yield

$$\frac{1}{2}\frac{d}{dt}\|\nabla^k v\|_{L^2}^2+\|\nabla\nabla^k v\|_{L^2}^2=-\int_{\mathbb{R}^n}\nabla^k(v\cdot\nabla v)\cdot\nabla^k v dx+\int_{\mathbb{R}^n}\nabla^k(\rho(u-v))\cdot\nabla^k v dx. \tag{57}$$

Then for $1\leq k\leq s+1$ that

$$\frac{d}{dt}\|\nabla^k v\|_{L^2}^2+\|\nabla\nabla^k v\|_{L^2}^2\leq C\delta(1+t)^{-\frac{n}{2}}N(t)^2. \tag{58}$$

□

Proof of high-order derivatives of v

Proof.

Define the sets $X_2(t)$ and $X_2^c(t)$ as follows

$$X_2(t) = \Big\{\xi : |\xi| \leq 1\Big\},\ X_2^c(t) = \Big\{\xi : |\xi| > 1\Big\}. \tag{59}$$

Therefore,

$$\frac{d}{dt}\|\nabla^k v\|_{L^2}^2 + \int_{X_2^c(t)} |\xi|^{2k}|\hat{v}(t,\xi)|^2 d\xi \leq C\delta(1+t)^{-\frac{n}{2}} N(t)^2. \tag{60}$$

$$\begin{aligned}\frac{d}{dt}\|\nabla^k v\|_{L^2}^2 + \|\nabla^k v\|_{L^2}^2 &\leq \int_{X_2(t)} |\xi|^{2k}|\hat{v}(t,\xi)|^2 d\xi + C\delta(1+t)^{-\frac{n}{2}} N(t)^2 \\ &\leq \cdots \\ &\leq C\mathcal{I}_0^2(1+t)^{-\frac{n}{2}} + C\delta(1+t)^{-\frac{n}{2}} N(t)^2.\end{aligned} \tag{61}$$

Summing k from 1 to $s+1$ yields

$$\|\nabla v(t)\|_{H^s} \leq C(1+t)^{-\frac{n}{4}}(\mathcal{I}_0 + \delta^{\frac{1}{2}} N(t)). \tag{62}$$

Proof of high-order derivatives of u

Proof.

Applying ∇^k to (54), multiplying it by $\nabla^k u$, and integrating

$$\frac{1}{2}\frac{d}{dt}\|\nabla^k u\|_{L^2}^2 + \|\nabla^k u\|_{L^2}^2 = \int_{\mathbb{R}^n} \nabla^k(-u\cdot\nabla u)\cdot\nabla^k u dx + \int_{\mathbb{R}^n} \nabla^k u\cdot\nabla^k v dx. \quad (63)$$

Then

$$\frac{d}{dt}\|\nabla u\|_{H^s}^2 + \|\nabla u\|_{H^s}^2 \le C(1+t)^{-\frac{n}{2}}(\mathcal{I}_0 + \delta^{\frac{1}{2}}N(t))^2. \quad (64)$$

By Grönwall inequality, we have

$$\|\nabla u(t)\|_{H^s} \le C(1+t)^{-\frac{n}{4}}(\mathcal{I}_0 + \delta^{\frac{1}{2}}N(t)). \quad (65)$$

□

Weighted estimate of $\|\nabla u(\tau)\|_{H^s}$

Proposition 3

Assume the conditions in Theorem 2.1 hold. Let $(\rho, u, v)(t, x)$ *be the classical solution of the system* (118)-(2) *satisfying the a priori assumption* (27). *Then, it holds for* $t \geq 0$ *that*

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla u(\tau)\|_{H^s}^2 d\tau \leq C(\mathcal{I}_0^2 + \mathcal{I}_0^3), \tag{66}$$

where the positive constant C *is independet of time.*

Proof of Proposition 3

Proof.

Multiplying (36) by $(1+t)^{9/8}$ gives

$$\frac{d}{dt}\Big\{(1+t)^{\frac{9}{8}}E(t)\Big\}+(1+t)^{\frac{9}{8}}\|\nabla v\|_{L^2}^2\le\frac{9}{8}(1+t)^{\frac{1}{8}}E(t). \tag{67}$$

It follows from Proposition 1 that

$$E(t)\le C\mathcal{I}_0^2(1+t)^{-\frac{n}{2}}, \tag{68}$$

$$\int_0^t(1+\tau)^{\frac{9}{8}}\|\nabla v(\tau)\|_{L^2}^2d\tau\le C\mathcal{I}_0^2\int_0^t(1+\tau)^{-\frac{n}{2}+\frac{1}{8}}d\tau+E_0\le C\mathcal{I}_0^2. \tag{69}$$

Then

$$\begin{aligned}&\int_0^t(1+\tau)^{\frac{9}{8}}\|\nabla u(\tau)\|_{L^2}^2d\tau\\&\le C\mathcal{I}_0^3\int_0^t(1+\tau)^{-\frac{3n}{4}+\frac{9}{8}}d\tau+\int_0^t(1+\tau)^{\frac{9}{8}}\|\nabla v(\tau)\|_{L^2}^2d\tau+\cdots\\&\le C(\mathcal{I}_0^2+\mathcal{I}_0^3).\end{aligned} \tag{70}$$

Proof of Proposition 3

Proof.

Let (54)-(118)$_3$, we obtain a new system for $(u-v)$

$$(u-v)_t + u - v = -u \cdot \nabla u + v \cdot \nabla v + \nabla P - \Delta v - \rho(u-v). \tag{71}$$

Multiply (71) by $(u-v)$ and integrate over $\mathbb{R}^n$

$$\frac{1}{2}\frac{d}{dt}\|u-v\|_{L^2}^2 + \|u-v\|_{L^2}^2 = \int_{\mathbb{R}^n} (\cdots) \cdot (u-v) dx. \tag{72}$$

By using the fact that

$$\|\nabla P\|_{L^2} \leq C\|v\|_{L^\infty}\|\nabla v\|_{L^2} + C\|\rho\|_{L^\infty}\|u-v\|_{L^2}.$$

Then, it holds

$$\frac{d}{dt}\|u-v\|_{L^2}^2 + \|u-v\|_{L^2}^2 \leq C\mathcal{I}_0^3(1+t)^{-\frac{3n}{4}} + 3\|\nabla v\|_{L^2}^2 + \|\nabla u\|_{L^2}^2. \tag{73}$$

□

Proof.

One has

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|(u-v)(\tau)\|_{L^2}^2 d\tau \le C(\mathcal{I}_0^2 + \mathcal{I}_0^3). \tag{74}$$

Then to estimate $\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla^2 v(\tau)\|_{L^2}^2 d\tau$. Applying ∇ to $(118)_3$ and taking the L^2 inner product with ∇v in $\mathbb{R}^n$ yield

$$\frac{1}{2}\frac{d}{dt}\|\nabla v\|_{L^2}^2 + \|\nabla^2 v\|_{L^2}^2 = -\int_{\mathbb{R}^n} \nabla(v\cdot\nabla v)\cdot\nabla v dx + \int_{\mathbb{R}^n} \nabla(\rho(u-v))\cdot\nabla v dx. \tag{75}$$

Integration by parts, we have

$$\left|\int_{\mathbb{R}^n} \nabla(\rho(u-v))\cdot\nabla v dx\right| \le C\|\rho\|_{L^\infty}\|u-v\|_{L^2}\|\nabla^2 v\|_{L^2} \le C\delta\|u-v\|_{L^2}^2 + \frac{1}{2}\|\nabla^2 v\|_{L^2}^2 \tag{76}$$

Then, it holds

$$\frac{d}{dt}\|\nabla v\|_{L^2}^2 + \|\nabla^2 v\|_{L^2}^2 \le C\mathcal{I}_0^3 (1+t)^{-\frac{3n}{4}} + C\delta\|u-v\|_{L^2}^2. \tag{77}$$

Proof of Proposition 3

Proof.

Multiplying (77) by $(1+t)^{9/8}$, intergrating it with time give

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla^2 v(\tau)\|_{L^2}^2 d\tau \leq C(\mathcal{I}_0^2 + \mathcal{I}_0^3). \tag{78}$$

Similar,

$$\begin{aligned} \int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla^2 u(\tau)\|_{L^2}^2 d\tau &\leq \cdots \\ &\leq C(\mathcal{I}_0^2 + \mathcal{I}_0^3). \end{aligned} \tag{79}$$

Repeat the above process and use induction method,

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla u(\tau)\|_{H^s}^2 d\tau \leq C(\mathcal{I}_0^2 + \mathcal{I}_0^3). \tag{80}$$

□

Closed A prior Estimates

Proposition 4

Assume the conditions in Theorem 2.1 hold. Let $(\rho, u, v)(t, x)$ *be the classical solution of the system* (118)-(2) *satisfying the a priori assumption* (27). *Then, it holds for* $t \geq 0$ *that*

$$\begin{aligned} &\|\rho(t)\|_{H^s}^2 + \|u(t)\|_{H^{s+2}}^2 + \|v(t)\|_{H^{s+1}}^2 + \\ &\int_0^t \left(\|\nabla u(\tau)\|_{H^{s+1}}^2 + \|\nabla v(\tau)\|_{H^{s+1}}^2\right) d\tau \leq C\delta_0^2, \end{aligned} \tag{81}$$

and

$$\rho(t, x) \geq \rho_0(X(0, x))e^{-C(\mathcal{I}_0^2 + \mathcal{I}_0^3)^{\frac{1}{2}}} > 0, \tag{82}$$

where the positive constant C *is independent of time.*

Proof.

Multiplying (118) by ρ and integrating over $\mathbb{R}^n$, we have

$$\frac{1}{2}\frac{d}{dt}\|\rho\|_{L^2}^2 \leq \left|\int_{\mathbb{R}^n} \rho \mathrm{div}(\rho u)dx\right| \leq \cdots \leq \|\mathrm{div} u\|_{L^\infty}\|\rho\|_{L^2}^2 \leq C\|\nabla^2 u\|_{H^{s-2}}\|\rho\|_{L^2}^2. \tag{83}$$

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\|\nabla\rho\|_{H^{s-1}}^2 &\leq C\|\nabla\rho\|_{H^{s-1}}(\|\nabla^2\rho\|_{H^{s-2}}\|\nabla u\|_{H^{s-1}} + \|\nabla^2 u\|_{H^{s-2}}\|\nabla\rho\|_{H^{s-1}}) + \cdots \\ &\leq C\|\nabla u\|_{H^s}\|\nabla\rho\|_{H^{s-1}}^2,\end{aligned}$$

Then

$$\frac{d}{dt}\|\rho\|_{H^s}^2 \leq C\|\nabla u\|_{H^s}\|\rho\|_{H^s}^2. \tag{84}$$

In terms of Proposition 3, it holds

$$\begin{aligned}\|\rho\|_{H^s}^2 &\leq \|\rho_0\|_{H^s}^2 \exp\left\{C\int_0^t \|\nabla u(\tau)\|_{H^s}d\tau\right\} \\ &\leq \|\rho_0\|_{H^s}^2 \exp\left\{C\int_0^t (1+\tau)^{-\frac{9}{16}}(1+\tau)^{\frac{9}{16}}\|\nabla u(\tau)\|_{H^s}d\tau\right\} \\ &\leq \|\rho_0\|_{H^s}^2 e^{C(\mathcal{I}_0^2+\mathcal{I}_0^3)^{\frac{1}{2}}} \leq C\delta_0^2.\end{aligned} \tag{85}$$

Define a backward characteristic $X(t,x)$ satisfies

$$\partial_\ell X(\ell,x) = u(\ell, X(\ell,x)) \text{ with } X(t,x) = x. \tag{86}$$

A straightforward calculation yields

$$\rho(t,x) = \rho_0(X(0,x)) \exp\Big\{ -\int_0^t \mathrm{div} u(\tau, X(\tau,x)) d\tau \Big\}. \tag{87}$$

By Proposition 3, we deduce that

$$\begin{aligned}
\Big| \int_0^t \mathrm{div} u(\tau, X(\tau,x)) d\tau \Big| &\leq C \int_0^t \|\nabla u(\tau)\|_{L^\infty} d\tau \\
&\leq C\Big(\int_0^t (1+\tau)^{-\frac{9}{8}} d\tau\Big)^{\frac{1}{2}} \Big(\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla u(\tau)\|_{H^s}^2 d\tau\Big)^{\frac{1}{2}} \\
&\leq C(\mathcal{I}_0^2 + \mathcal{I}_0^3)^{\frac{1}{2}}.
\end{aligned}$$

Since the initial density $\rho_0 > 0$ for $x \in \mathbb{R}^n$, it gives rise to

$$\rho(t,x) \geq \rho_0(X(0,x)) e^{-C(\mathcal{I}_0^2 + \mathcal{I}_0^3)^{\frac{1}{2}}} > 0. \tag{88}$$

Proof of Theorem 2.1. By Proposition 4, we obtain

$$\begin{aligned} &\|\rho(t)\|_{H^s}^2 + \|u(t)\|_{H^{s+2}}^2 + \|v(t)\|_{H^{s+1}}^2 \\ &+ \int_0^t \left(\|\nabla u(\tau)\|_{H^{s+1}}^2 + \|\nabla v(\tau)\|_{H^{s+1}}^2 \right) d\tau \le C\delta_0^2. \end{aligned} \tag{89}$$

Choosing the initial data δ_0 sufficiently small such that $C\delta_0^2 \le \frac{1}{4}\delta^2$, we obtain

$$\sup_{t\ge 0} \mathcal{Z}(t) = \sup_{t\ge 0} \left(\|\rho(t)\|_{H^s}^2 + \|u(t)\|_{H^{s+2}}^2 + \|v(t)\|_{H^{s+1}}^2 \right)^{\frac{1}{2}} \le \frac{1}{2}\delta, \tag{90}$$

which, together with (82), closes the a priori assumption (27).
By Proposition 1 and Proposition 2, it is easy to verify for $s \ge 2[\frac{n}{2}] + 1$ that

$$\|(u,v)(t)\|_{L^2} \le C\mathcal{I}_0(1+t)^{-\frac{n}{4}}, \quad \|\nabla(u,v)(t)\|_{H^s} \le C\mathcal{I}_0(1+t)^{-\frac{n}{4}}. \tag{91}$$

This completes the proof of Theorem 2.1.

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Proposition 5

Assume the same conditions in Theorem 2.1 hold. There exists a sufficiently small constant $\delta_0 > 0$ *such that if*

$$\|\rho_0\|_{H^s(\mathbb{R}^n)} + \|u_0\|_{H^{s+2}(\mathbb{R}^n)} + \|v_0\|_{H^{s+1}(\mathbb{R}^n)} + \|v_0\|_{L^1(\mathbb{R}^n)} \leq \delta_0, \tag{92}$$

and the Fourier transform of the initial velocity $\hat{v}_0(\xi)$ *satisfy*

$$\inf_{|\xi|\leq 1} |\hat{v}_0(\xi)| \geq \delta_0^{\frac{3}{2}}, \tag{93}$$

then it holds for large time that

$$\|v(t)\|_{L^2} \geq \frac{1}{2}\delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}}. \tag{94}$$

Proof of Proposition 5

Since the assumption of $\|v_0\|_{L^1}$ in (92) is also sufficiently small, according to the result proved by Proposition 3.1 [Choi-Jung-2021-JMFM] , we have for any $\alpha \in (0, n/4)$,

$$E(t)(1+t)^{2\alpha} + \int_0^t (1+\tau)^{2\alpha} D(\tau)d\tau \leq C(E(0) + \|v_0\|_{L^1}^2) \leq C\delta_0^2, \quad (95)$$

which gives

$$E(t) = \int_{\mathbb{R}^n} \rho|u|^2 dx + \|v\|_{L^2}^2 \leq C\delta_0^2(1+t)^{-2\alpha}, \quad \forall\ 0 < \alpha < \frac{n}{4}. \quad (96)$$

Choosing $\alpha = \frac{n}{4} - \frac{1}{16}$, we have

$$E(t) \leq C\delta_0^2(1+t)^{-\frac{n}{2}+\frac{1}{8}}, \quad \|v(t)\|_{L^2} \leq C\delta_0(1+t)^{-\frac{n}{4}+\frac{1}{16}}. \quad (97)$$

Proof of Proposition 5

We follow a similar method developed in Proposition 3 to show

$$\|\nabla(u,v)(t)\|_{H^s} \le C\delta_0(1+t)^{-\frac{n}{4}+\frac{1}{16}}. \tag{98}$$

Then we have

$$\frac{d}{dt}\Big\{(1+t)^{\frac{9}{8}}E(t)\Big\} + (1+t)^{\frac{9}{8}}\|\nabla v\|_{L^2}^2 \le \frac{9}{8}(1+t)^{\frac{1}{8}}E(t) \le C\delta_0^2(1+t)^{-\frac{n}{2}+\frac{1}{4}}. \tag{99}$$

Noting that $n \ge 3$, therefore integrating (99) with time yields

$$\int_0^t (1+\tau)^{\frac{9}{8}}\|\nabla v(\tau)\|_{L^2}^2 d\tau \le C\delta_0^2\int_0^t (1+\tau)^{-\frac{n}{2}+\frac{1}{4}}d\tau + E_0 \le C\delta_0^2. \tag{100}$$

Similar as Proposition (2),

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|\nabla u(\tau)\|_{L^2}^2 d\tau \leq \cdots \leq C(\delta_0^2 + \delta_0^4). \tag{101}$$

$$\int_0^t (1+\tau)^{\frac{9}{8}} \|(u-v)(\tau)\|_{L^2}^2 d\tau \leq \cdots \leq C\delta_0^2. \tag{102}$$

Denote $q = v - w$,

$$\begin{cases} w_t - \Delta w = 0, \\ w(0,x) = v_0(x), \quad \mathrm{div} v_0 = 0. \end{cases} \tag{103}$$

and

$$\begin{cases} q_t + v \cdot \nabla v + \nabla P = \Delta q + \rho(u - v), \\ \mathrm{div} q = 0, \quad q(0,x) = 0. \end{cases} \tag{104}$$

Proof of Proposition 5

$$
\begin{aligned}
&\frac{1}{2}\frac{d}{dt}(\int_{\mathbb{R}^n}\rho|u|^2dx+\|q\|_{L^2}^2)+\int_{\mathbb{R}^n}\rho|u-v|^2dx+\|\nabla q\|_{L^2}^2\\
&=-\int_{\mathbb{R}^n}w\cdot\rho(u-v)dx.
\end{aligned}
\tag{105}
$$

In terms of Cauchy inequality and Sobolev inequality, one has

$$
\begin{aligned}
&\frac{1}{2}\frac{d}{dt}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx+\|q\|_{L^2}^2\Big)+\int_{\mathbb{R}^n}\rho|u-v|^2dx+\|\nabla q\|_{L^2}^2\\
&\le\|w\|_{L^\infty}\int_{\mathbb{R}^n}\rho|u-v|dx\\
&\le\cdots\\
&\le\frac{1}{2}\|w\|_{L^\infty}^2\|\rho_0\|_{L^1}+\frac{1}{2}\int_{\mathbb{R}^n}\rho|u-v|^2dx\\
&\le C\mathcal{I}_0^3(1+t)^{-n}+\frac{1}{2}\int_{\mathbb{R}^n}\rho|u-v|^2dx,
\end{aligned}
\tag{106}
$$

Proof of Proposition 5

With the help of the Plancherel Theorem, we verify

$$\frac{1}{2}\frac{d}{dt}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx+\|q\|_{L^2}^2\Big)+\int_{X_3(t)}|\xi|^2|\hat{q}(t,\xi)|^2d\xi+\int_{X_3^c(t)}|\xi|^2|\hat{q}(t,\xi)|^2d\xi$$
$$+\frac{1}{2}\int_{\mathbb{R}^n}\rho|u-v|^2dx\le C\mathcal{I}_0^3(1+t)^{-n},$$

where $X_3(t)$ and $X_3^c(t)$ are defined by

$$X_3(t)=\Big\{\xi:|\xi|\le\Big(\frac{2n}{t+4n}\Big)^{\frac{1}{2}}\Big\},\quad X_3^c(t)=\Big\{\xi:|\xi|>\Big(\frac{2n}{t+4n}\Big)^{\frac{1}{2}}\Big\}. \tag{107}$$

$$\begin{aligned}&\frac{1}{2}\frac{d}{dt}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx+\|q\|_{L^2}^2\Big)+\frac{n}{2(t+4n)}\Big(\int_{\mathbb{R}^n}\rho|u|^2dx+\|q\|_{L^2}^2\Big)\\&\le\frac{2n}{t+4n}\int_{X_3(t)}|\hat{q}(t,\xi)|^2d\xi+C\mathcal{I}_0^3(1+t)^{-n}.\end{aligned} \tag{108}$$

To calculate $\int_{X_3(t)}|\hat{q}(t,\xi)|^2d\xi$.

$$q_t+v\cdot\nabla v+\nabla P=\Delta q+\rho(u-v).$$

Proof of Proposition 5

$$\begin{cases} q_t - \Delta q = F, \\ q(0,x) = 0. \end{cases} \tag{109}$$

We have

$$\hat{q}(t,\xi) = \int_0^t e^{-|\xi|^2(t-\tau)} \hat{F}(\tau,\xi) d\tau.$$

$$|\hat{F}(\tau,\xi)| \leq |\xi| \|v\|_{L^2}^2 + \|\rho(u-v)\|_{L^1} \leq C|\xi| \mathcal{I}_0^2 (1+\tau)^{-\frac{n}{2}} + \|\rho(u-v)\|_{L^1}.$$

$$\begin{aligned} \int_0^t \|\rho(u-v)(\tau)\|_{L^1} d\tau &\leq \int_0^t \|\rho(\tau)\|_{L^2} \|(u-v)(\tau)\|_{L^2} d\tau \\ &\leq C\delta_0 \Big(\int_0^t (1+\tau)^{-\frac{9}{8}} d\tau\Big)^{\frac{1}{2}} \Big(\int_0^t (1+\tau)^{\frac{9}{8}} \|(u-v)(\tau)\|_{L^2}^2 d\tau\Big)^{\frac{1}{2}} \\ &\leq C\delta_0^2. \end{aligned}$$

Hence, we get

$$|\hat{q}(t,\xi)| \leq C|\xi| \mathcal{I}_0^2 e^{-\frac{1}{2}|\xi|^2 t} + C|\xi|^{-1} \mathcal{I}_0^2 (1+t)^{-\frac{n}{2}} + C\delta_0^2. \tag{110}$$

Define the energy $H(t)$ as follows

$$H(t) = \int_{\mathbb{R}^n} \rho|u|^2 dx + \|q\|_{L^2}^2, \quad H(0) = \int_{\mathbb{R}^n} \rho_0|u_0|^2 dx.$$

It holds

$$\begin{aligned} &\frac{d}{dt}H(t) + \frac{n}{t+4n}H(t) \\ &\leq C\mathcal{I}_0^4(1+t)^{-\frac{n+4}{2}} + C\delta_0^4(1+t)^{-\frac{n+2}{2}} + C\mathcal{I}_0^3(1+t)^{-n}. \end{aligned} \tag{111}$$

Then, multiplying (111) by $(t+4n)^n$ and integrating the resulting equation with time gives

$$\begin{aligned} H(t) \leq\ & C\mathcal{I}_0^4(1+t)^{-\frac{n+2}{2}} + H(0)(1+t)^{-n} \\ & + C\delta_0^4(1+t)^{-\frac{n}{2}} + C\mathcal{I}_0^3(1+t)^{-n+1}. \end{aligned}$$

Hence, we obtain the following decay rate for large time that

$$\|q(t)\|_{L^2} \leq \cdots \leq C\delta_0^2(1+t)^{-\frac{n}{4}}. \tag{112}$$

Proof of Proposition 5

In terms of Lemma 3, (112) and the smallness of δ_0, we can prove that for large time that

$$\begin{aligned} \|v\|_{L^2} &\geq \|w\|_{L^2} - \|q\|_{L^2} \\ &\geq \delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}} - C\delta_0^2(1+t)^{-\frac{n}{4}} \\ &\geq \frac{1}{2}\delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}}. \end{aligned} \tag{113}$$

Proof of Theorem 2.2

By Proposition 1, we have the upper bound time decay rate of the velocity,

$$\|v(t)\|_{L^2} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}. \tag{114}$$

According to Proposition 5, we obtain the lower bound decay rate of velocity for large time

$$\|v(t)\|_{L^2} \geq \frac{1}{2}\delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}}. \tag{115}$$

Thus, we combine them together to prove

$$\frac{1}{2}\delta_0^{\frac{3}{2}}(1+t)^{-\frac{n}{4}} \leq \|v(t)\|_{L^2} \leq C\mathcal{I}_0(1+t)^{-\frac{n}{4}}. \tag{116}$$

This completes the proof of Theorem 2.2.

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Furture works

- Global well-posedness and large-time behavior of classical solutions to the Vlasov-Poisson-Navier-Stokes

$$\begin{cases} \partial_t f + \xi \cdot \nabla_x f + \mathrm{div}_\xi((v - \xi - \nabla_x U)f) = -\alpha \mathrm{div}_\xi((u_f - \xi)f) + \sigma \Delta_\xi f, \\ -\Delta_x U = \rho - c, \\ \partial_t v + v \cdot \nabla_x v + \nabla_x P = \Delta_x v + \displaystyle\int_{\mathbb{R}^n} (\xi - v) f d\xi, \\ \mathrm{div}_x v = 0. \end{cases} \tag{117}$$

- Global well-posedness and large-time behavior of classical solutions to the Euler- Poisson-Navier-Stokes system

$$\begin{cases} \partial_t \rho + \mathrm{div}(\rho u) = 0, \\ \partial_t(\rho u) + \mathrm{div}\big(\rho u \otimes u\big) + \nabla \rho = -\rho(u - v) - \rho \nabla U, \\ -\Delta U = \rho - c, \\ \partial_t v + v \cdot \nabla v + \nabla P = \Delta v + \rho(u - v), \\ \mathrm{div} v = 0. \end{cases} \tag{118}$$

Thank you!